

INFRASTRUCTURE RESOURCE REPORT — PHASE 1

Infrastructure Report — Phase 1

The Archive MVP

v2.0 · Richard Uzor — May 2026 · Devstrike Digital Limited

01 — PHASE 1 INFRASTRUCTURE PRINCIPLE

Phase 1 runs almost entirely on free tiers. The only recurring cost is Plausible Analytics at ₦14,400/month. Total burn is under ₦25,000/month. No engineering time should be spent on infrastructure optimisation in Phase 1.

02 — PHASE 1 RESOURCES

Frontend Hosting *Vercel Hobby*

Tier: Free

Estimated Cost: ₦0/month

Hosts the Next.js application. Serverless functions for API routes. GitHub integration for zero-config deploys. 100GB bandwidth and 6,000 build minutes per month on free tier.

Scaling trigger: Upgrade to Pro when monthly bandwidth > 80GB or team grows beyond 1 developer.

Backend API + Database *Railway Starter*

Tier: Free (\$5 credit/month)

Estimated Cost: ₦0-₦2,000/month

Node.js API in a Docker container + PostgreSQL 15 managed database. Auto-restart, environment variables, GitHub deploy integration. Free tier covers Phase 1 traffic comfortably.

Scaling trigger: Upgrade when API CPU consistently > 70% or free credit runs out before month end.

Object Storage + CDN *Cloudflare R2 + Free CDN*

Tier: Pay-as-you-go

Estimated Cost: ₦2,000-₦4,000/month

Stores pastor/church images and generated quote card PNG files. R2 has zero egress fees — no cost for serving images to users. Cloudflare CDN adds edge caching for free.

Scaling trigger: Costs scale linearly at \$0.015/GB stored + \$0.36/million Class A operations.

Transactional Email *Resend*

Tier: Free (3,000 emails/month)

Estimated Cost: ₦0/month

Sends account verification emails, Q&A notifications (email to pastor if contact on file), report threshold alerts. React Email templates for consistent branding.

Scaling trigger: Upgrade to paid when monthly sends > 2,400. Typically at ~500 MAU with notifications enabled.

Analytics *Plausible Analytics*

Tier: Starter

Estimated Cost: ₦14,400/month (~\$9)

Privacy-first analytics. No cookies, no GDPR consent banner. Tracks: pageviews, referral sources (WhatsApp share traffic is the key Phase 1 signal), popular messages and pastors.

Scaling trigger: No upgrade needed in Phase 1. Plausible Business (\$19/mo) only if custom funnels needed.

Error Tracking *Sentry*

Tier: Free tier

Estimated Cost: ₦0/month

Frontend React error boundaries + backend Express error handler. Full stack traces, user context. Alerts on new error types.

Scaling trigger: Upgrade when monthly errors > 5,000.

03 — PHASE 1 COST SUMMARY

Item	Monthly Cost
Vercel Hobby	₦0
Railway Starter	₦0–₦2,000
Cloudflare R2	₦2,000–₦4,000
Resend	₦0
Plausible	₦14,400
Sentry	₦0
Domain (amortised)	₦1,500
Misc	₦2,000
TOTAL	₦19,900–₦23,900/month